

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II
WEEK 12: LECTURE NOTE

TECHNIQUES OF INTEGRATION

Integration by Parts

Theorem 1. *If u and v are functions of x and have continuous derivatives, then*

$$\int u dv = uv - \int v du.$$

GUIDELINES FOR INTEGRATION BY PARTS

1. Try letting dv be the most complicated portion of the integrand that fits a basic integration rule. Then u will be the remaining factor(s) of the integrand.
2. Try letting u be the portion of the integrand whose derivative is a function simpler than u . Then dv will be the remaining factor(s) of the integrand.

Note that dv always includes the dx of the original integrand.

Example 2 (Integration by Parts). Find $\int x e^x dx$.

Solution. In class

Example 3 (Integration by Parts). Find $\int x^2 \ln x dx$.

Solution. In class.

Example 4 (Repeated Use of Integration by Parts). Find $\int x^2 \sin x dx$.

Solution. In class.

Example 5 (Integration by Parts). Find $\int \sec^3 x dx$.

Solution. In class. **SUMMARY: COMMON INTEGRALS USING INTEGRATION BY PARTS**

1. For integrals of the form

$$\int x^n e^{ax} dx, \quad \int x^n \sin ax dx, \quad \text{or} \quad \int x^n \cos ax dx,$$

let $u = x^n$ and let $dv = e^{ax} dx$, $dv = \sin ax dx$, or $dv = \cos ax dx$.

2. For integrals of the form

$$\int x^n \ln x dx$$

let $u = \ln x$ and let $dv = x^n dx$.

3. For integrals of the form

$$\int e^{ax} \sin bx \, dx \quad \text{or} \quad \int e^{ax} \cos bx \, dx$$

let $u = \sin bx$ or $u = \cos bx$ and $dv = e^{ax} dx$.

Example 6. *Evaluating Definite Integrals by Parts*

Find $\int_0^{\frac{\pi}{4}} x \cos 2x dx$.

Solution.. In class

Integration by the Method of Partial Fractions

A procedure for decomposing a rational function into simpler rational functions to which we can apply the basic integration formulas is called **the method of partial fractions**. The general description of the method is given below.

DECOMPOSITION OF $\frac{N(x)}{D(x)}$ INTO PARTIAL FRACTIONS

1. **Divide when improper:** When $\frac{N(x)}{D(x)}$ is an improper fraction (that is, when the degree of the numerator is greater than or equal to the degree of the denominator), divide the denominator into the numerator to obtain

$$\frac{N(x)}{D(x)} = (\text{a polynomial}) + \frac{N_1(x)}{D(x)}$$

where the degree of $N_1(x)$ is less than the degree of $D(x)$. Then apply steps 2, 3, and 4 to the proper rational expression $\frac{N_1(x)}{D(x)}$.

2. **Factor denominator:** Completely factor the denominator into factors of the form

$$(px + q)^m \quad \text{and} \quad (ax^2 + bx + c)^n$$

where $ax^2 + bx + c$ is irreducible.

3. **Linear factors:** For each factor of the form $(px + q)^m$, the partial fraction decomposition must include the following sum of m fractions.

$$\frac{A_1}{(px + q)} + \frac{A_2}{(px + q)^2} + \cdots + \frac{A_m}{(px + q)^m}$$

4. **Quadratic factors:** For each factor of the form $(ax^2 + bx + c)^n$ the partial fraction decomposition must include the following sum of n fractions.

$$\frac{B_1x + C_1}{(ax^2 + bx + c)} + \frac{B_2x + C_2}{(ax^2 + bx + c)^2} + \cdots + \frac{B_nx + C_n}{(ax^2 + bx + c)^n}$$

Example 7 (DISTINCT LINEAR FACTORS). *Find*

$$\int \frac{1}{x^2 - 5x + 6} dx.$$

Solution. In class.

Example 8 (REPEATED LINEAR FACTORS). *Find*

$$\int \frac{5x^2 + 20x + 6}{x^3 + 2x^2 + x} dx.$$

Solution. In class.

Example 9 (DISTINCT LINEAR AND QUADRATIC FACTORS). *Find*

$$\int \frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)} dx.$$

Solution. In class